AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 17 · STUDENT LESSON

Prediction Is Not Fact

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

Save the prediction. Check the separate evidence.

Trace ramp and drying records, preserve mismatches, and report assessed cases alongside unresolved observations.

Keep one habit: Link the case → save the output → apply the observation rule without rewriting either record.

MAIN LESSON · SECTION 1

What checks a prediction?

> Keep the model output, later evidence and comparison status separate for the same case.

Start with a correctly computed wrong prediction

A rule can produce exactly the output its instructions require while that prediction disagrees with a later observation. Save both records so you can check what happened instead of rewriting the prediction.

Retrieve the unresolved-record question

A model predicts far, but the distance record is unreadable. Can this case be called correct? What **evidence** would let you check it?

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

By the end, you can...

- I can separate a recorded **prediction** from the **observation** used to check it.
- I can compare matching cases using a stated **outcome rule** and preserve unresolved **evidence**.
- I can report checked results without counting missing **observation**s as successes or failures.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Boundaries and 'Not Sure'. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Build a Human Classifier. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Repeating Efficiently. Give an example and a mistake that the idea helps you avoid.

A **prediction** is a fallible **output** supported by clues.
</details>



LEARNING TARGET On Monday a team predicts 'rain on Tuesday'. On Tuesday they count a damp playground as success, although sprinklers ran. Write an **observation** rule that could have been agreed before Tuesday.

Learner name:

Readiness for the connected explanation

This week’s end goal

By the end, I can keep predicted and observed outcomes separate and audit a disagreement.

MAIN LESSON · SECTION 2

Use separate prediction and observation rules

> Ramp height generates a prediction; later travel distance provides different evidence.

State the comparison protocol

A **prediction** states what a model **output**s. An **observation** records what was measured or happened. To check a **prediction**, keep both records and compare the same case under the same **outcome rule**.

Writing a **prediction** into the **observation** column would make the check circular.

Read Ramp1 and its outcome rule

Use a fictional ramp investigation. Before each trial, Ramp1 predicts far for a ramp height of at least 6 cm and near otherwise. Its allowed heights are whole centimetres from 1 through 10.

After a toy rolls, a learner measures its travel distance. The **outcome rule** is far for a distance of at least 4 floor units and near otherwise. Ramp height and travel distance are different quantities.

Keep the fictional records linked by trial ID

The values in this lesson are invented practice records, not a physical law about ramps. Ramp1 is a fixed paper model for learning how to check **prediction**s.

Each trial has an ID so that its height, **prediction** and later measurement stay together. Keep the model version and record the **prediction** before opening the **observation** card.

Explain the two measurements

- Which quantity is the prediction input, and which later quantity determines the observed category? Explain why a trial ID, model version and saved pre-observation prediction are needed.

▼ Earlier wording, alternatives and picture transcripts

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



MYSTERY A rain **prediction** uses weather clues, but the later **observation** of rain or no rain is the **fact** that evaluates it.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first **prediction** and state the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

A closer explanation

A mismatch preserves two different records

> Use the original prediction even when later evidence disagrees.

Compare the three established ramp trials

Two records meet at a comparison

BEFORE THE TRIAL	AFTER THE TRIAL	COMPARISON
A: height 4 cm	Distance 3 units	Near matches near
B: height 6 cm	Distance 3 units	Far differs from near
C: height 8 cm	Distance 5 units	Far matches far

Ramp1 predicts from height: A near, B far, C far. The observation is travel distance. Apply the distance outcome rule separately, then compare categories for the same trial.

Explain B’s mismatch without editing its history

For B, the **input** 6 cm produces the **prediction** far. The later distance of 3 units produces the observed category near. This is a **mismatch**. Retain far as the original **prediction** and near as the observed category.

Changing the old **prediction** to near would hide the mistake rather than improve the original result.

Limit what A and C's matches establish

For A and C, the categories match. These two matches provide **evidence** about these trials. They do not show that every future **prediction** will be correct.

Even a correctly computed **prediction** can disagree with an **observation** because following the model rule and matching the world are different checks.

Explain one comparison

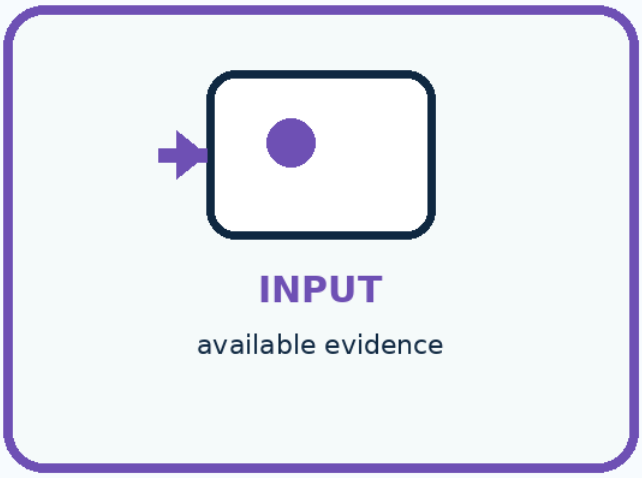
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

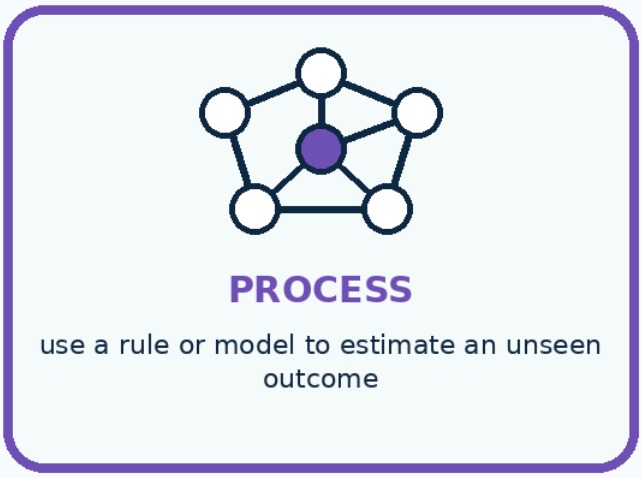
▼ Earlier wording, alternatives and picture transcripts

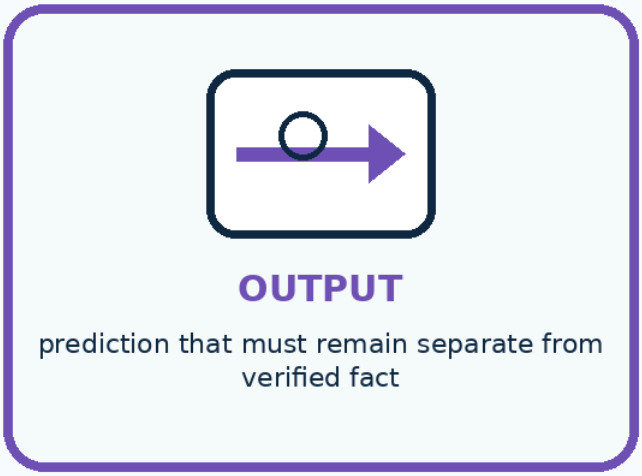
VISUAL **MECHANISM**

Input → Process → Output

Follow the information instead of imagining magic inside the machine.







INPUT available **evidence**

PROCESS use a rule or model to estimate an unseen outcome

OUTPUT prediction that must remain separate from verified **fact**

CORE IDEA A **prediction** is a fallible **output** supported by clues.

UNDER THE HOOD **Prediction** estimates a target that is unknown at decision time. **Evidence** supports the estimate; **ground truth** is the later or independently established outcome.

Treating **prediction**s as labels can create self-confirming records.

Draw a fresh example with all three stages:

A closer explanation

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Compare predictions, observations and evidence claims

> Ask what was recorded before the outcome and what evidence checks it afterwards.

Pictures paired with their evidence



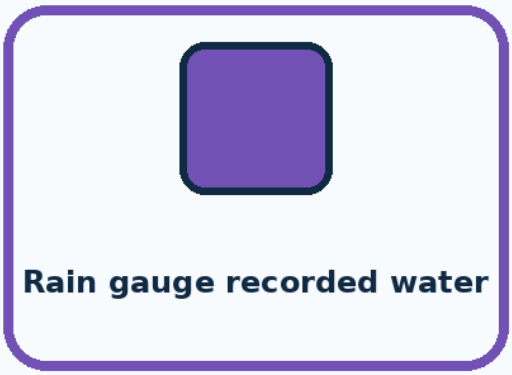
The case to investigate

CASE 1 — Rain may start Clue: forecast **output**. Meaning: **prediction**.



The case to investigate

CASE 2 — Dark clouds Clue: observed condition. Meaning: **evidence**.



The case to investigate

CASE 3 — Rain gauge recorded water Clue: later measurement. Meaning: **fact** for evaluation.



The case to investigate

CASE 4 — Animal label from model Clue: unverified **output**. Meaning: **prediction**, not **fact**.

Discuss the confusing pair

- Which two pictures are easiest to confuse? What **evidence** separates them?

Keep the limit in view

BOUNDARY Do not copy a model **prediction** into the answer column and later call it **ground truth**.

▼ Earlier wording, alternatives and picture transcripts

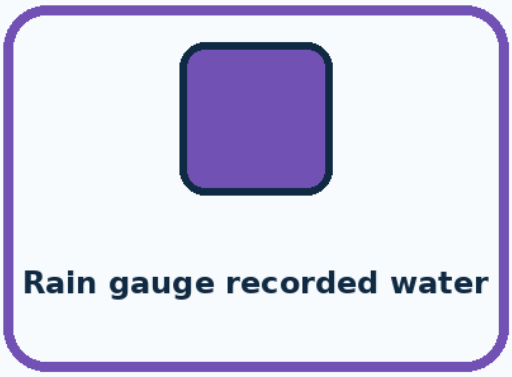
PICTURE GALLERY

Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.









CASE 1 — Rain may start Clue: forecast **output**. Meaning: **prediction**.

CASE 2 — Dark clouds Clue: observed condition. Meaning: **evidence**.

CASE 3 — Rain gauge recorded water Clue: later measurement. Meaning: **fact** for evaluation.

CASE 4 — Animal label from model Clue: unverified **output**. Meaning: **prediction**, not **fact**.

Change one thing in the pictured case

Change: Keep score 4 but measure exactly 8 minutes.

Prediction: The observed label becomes quick and agrees with the **prediction**.

Reason: Above 8 is strict: equality is quick under this policy.

The prediction input is not the observed outcome

> This separate drying demonstration uses a moisture-score model and a later time measurement.

The two rules and supplied trial

A fictional drying trial uses a fixed rule and later measures what actually happened.

Model D predicts slow when moisture score is at least 5; quick otherwise.

Observation policy: slow if drying takes above 8 minutes; quick otherwise. Trial T has score 4 and measured time 9 minutes.

Compare the two recorded quantities

AI ACADEMY / AI-1/17 · 03 · INSPECT THE LESSON MODEL

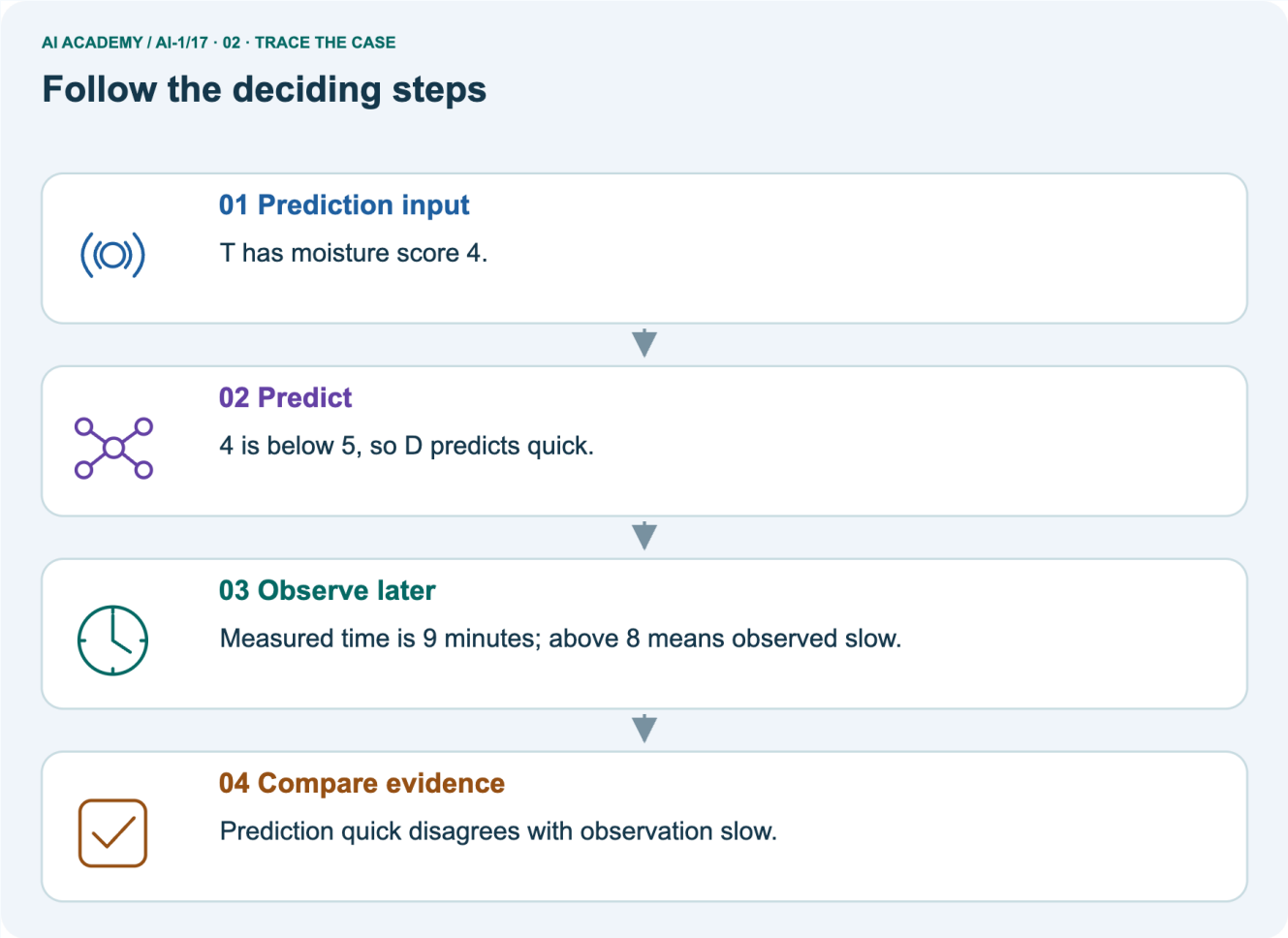
Two records meet at a comparison

Before the trial	After the trial	Comparison
A: height 4 cm	Distance 3 units	Near matches near
B: height 6 cm	Distance 3 units	Far differs from near
C: height 8 cm	Distance 5 units	Far matches far

Ramp1 predicts from height: A near, B far, C far. The observation is travel distance. Apply the distance outcome rule separately, then compare categories for the same trial.

Follow the labelled picture with the matching explanation. Open the vector diagram

Trace prediction and later comparison



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain what each record establishes

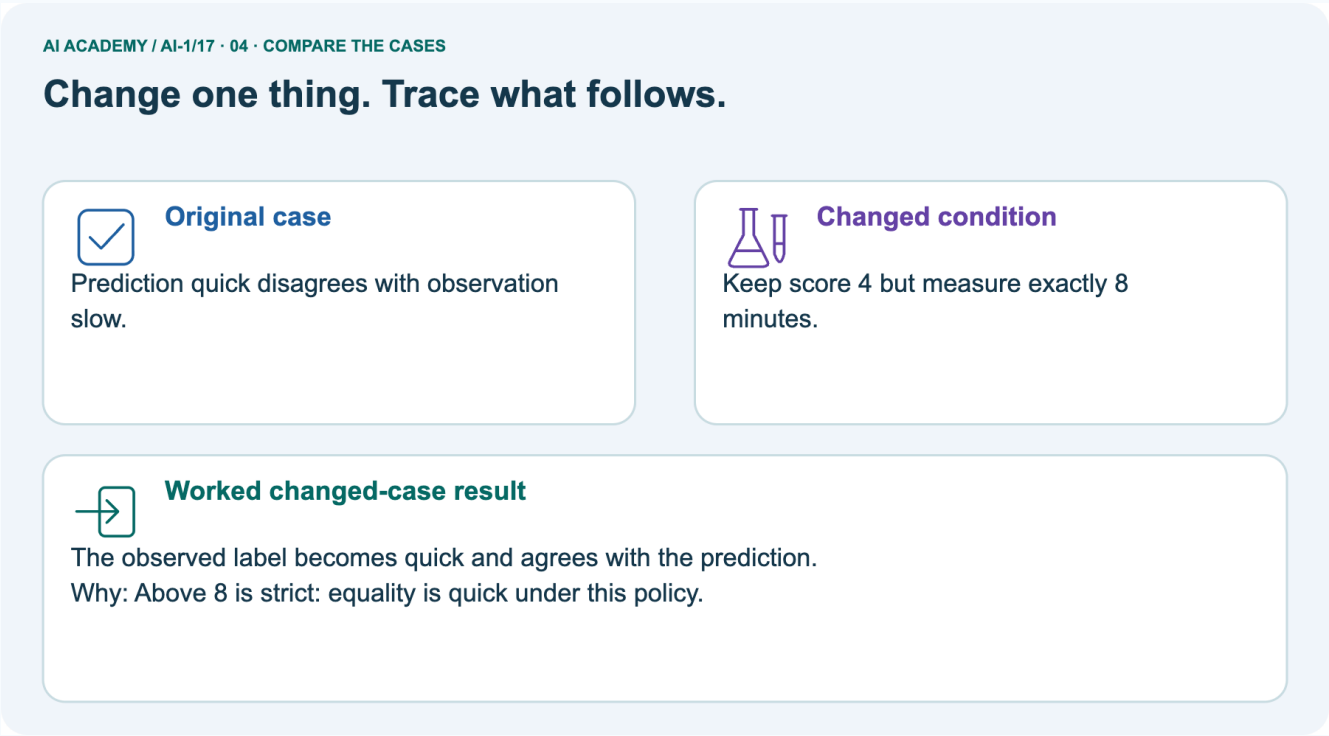
The moisture score generates a **prediction**; it is not the observed drying time. The later measurement supplies separate **evidence** under the stated **observation** policy. Save both values and their units.

Altering the recorded time to match D would destroy the evaluation rather than repair the model.

Predict the equality outcome

- Before checking, keep moisture score 4 and model D unchanged, but use a supplied measured drying time of exactly 8 minutes. Apply the strict above-8 outcome policy and compare with the saved model prediction.

▼ Check the changed observation



Follow the labelled picture with the matching explanation. Open the vector diagram

Limit a single match or mismatch

One failure does not prove every **prediction** is wrong; one agreement does not establish reliability on future trials.

Identify which field changes which category

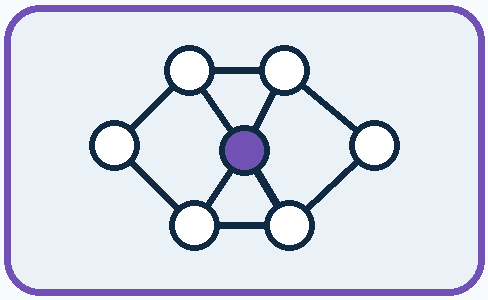
- Read the labelled picture: Which field changes the **observation** label, and which field changes the model **prediction**?

▼ **Earlier wording, alternatives and picture transcripts**

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



CASE A rain **prediction** uses weather clues, but the later **observation** of rain or no rain is the **fact** that evaluates it.

STEP 1 — OBSERVE available **evidence**

STEP 2 — TRACE use a rule or model to estimate an unseen outcome

STEP 3 — CONCLUDE **prediction** that must remain separate from verified **fact**

MODEL REASONING Mark model or rule **output**s as **prediction**s, observable clues as **evidence**, and independently checked outcomes as **fact**s.

A **prediction** becomes evaluated against a **fact**; it does not become a **fact** merely because time passes.

DO NOT OVERCLAIM Do not copy a model **prediction** into the answer column and later call it **ground truth**.

What new **evidence** could change this answer?

A closer explanation

Record	Evidence available	Status under the outcome rule
D: height 5 cm; predicts near	Distance missing	Unassessed; request the trial record
E: height 7 cm; predicts far	Two records say 3 and 4 units	Unassessed; investigate the conflict
F: height 6 cm; predicts far	Verified distance exactly 4 units	Match; at least 4 includes 4

For E, 3 units means near and 4 units means far. The disagreement can change the judgment. Check the trial ID, original measurement and recording procedure. Do not choose the distance that makes Ramp1 look better.

If the conflict cannot be resolved, retain both reports and mark the comparison **unassessed**.

An **observation** is **evidence**, not an automatic guarantee of truth. A wrong trial ID, unclear unit or copying error can make the comparison unreliable.

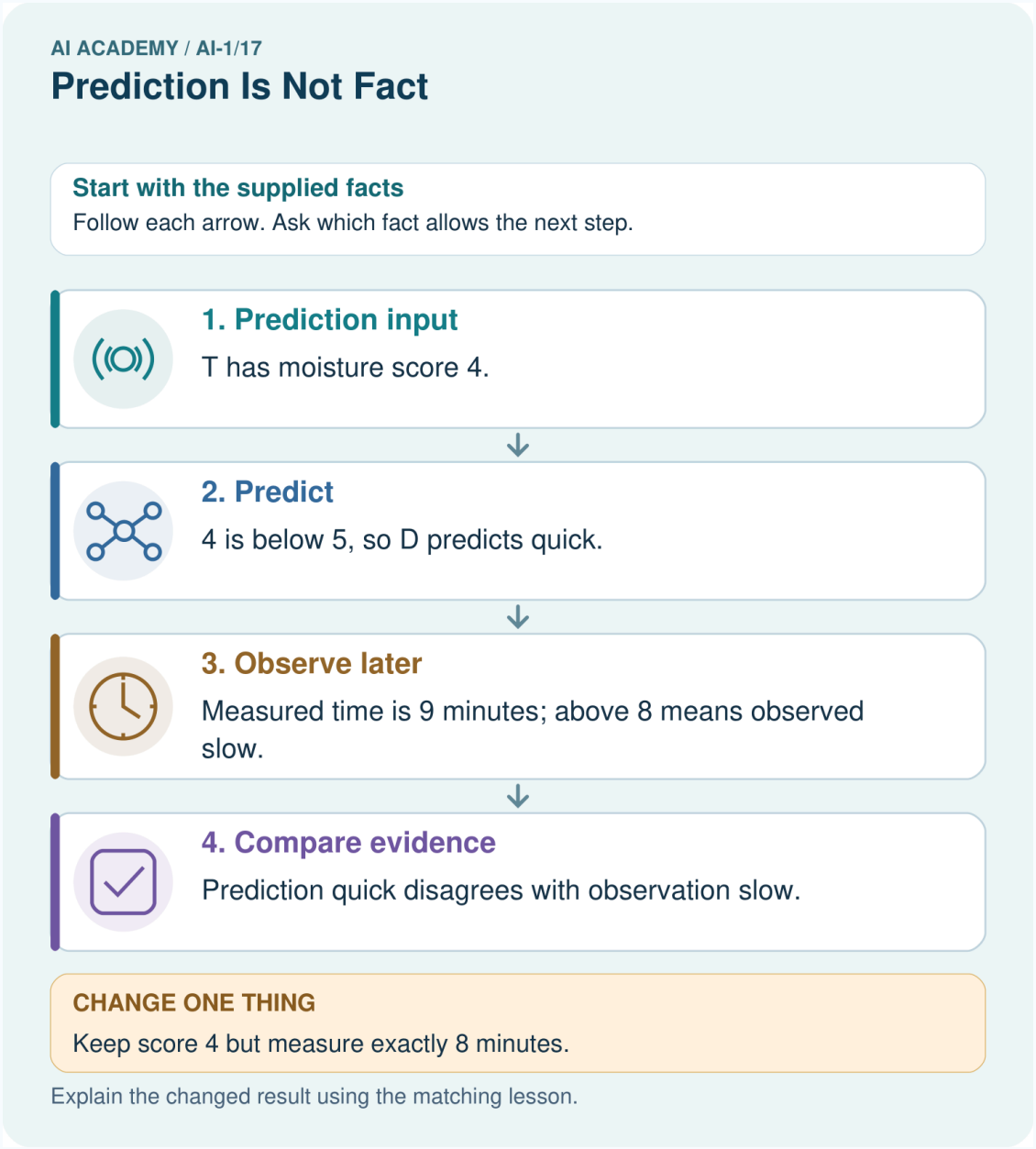
If a supported correction is made, retain the original entry, the correction, its source and the reason. Remeasuring a new trial does not reveal what happened in the old one.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first **mismatch**. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Prediction Is Not Fact

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Prediction input	T has moisture score 4.
2. Predict	4 is below 5, so D predicts quick.
3. Observe later	Measured time is 9 minutes; above 8 means observed slow.
4. Compare evidence	Prediction quick disagrees with observation slow.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

Change one thing: Keep score 4 but measure exactly 8 minutes. Predict the result before checking the explanation. What remains the same?

MAIN LESSON · SECTION 7

Investigate a mismatch without rewriting the old prediction

> Check IDs, input, calculation and outcome evidence before identifying the cause.

Choose the stage to inspect from the record

Start a **mismatch** investigation by checking the **input**, trial ID, **prediction** calculation and **observation** procedure. A **mismatch** alone does not identify which part caused the problem.

If all records are sound, this case is **evidence** that the model **prediction** did not match the outcome.

Keep a revised model in a new version

If you revise the model after seeing B, record a new version. A new **prediction** for B can help explain the change, but it is no longer a check on an unseen case.

Preserve Ramp1's result and use new reserved trials to evaluate the revised version, following Week 15's method.

Repair a rewritten perfect-score claim

- CLAIM A A confident **prediction** is a **fact**.
- CLAIM B **Evidence** and **prediction** are two words for the same thing.

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

> **A learner changes B's old prediction from far to near after reading its distance, then reports three matches out of three. Explain what is wrong with the report and how to preserve an honest record**

▼ **Earlier wording, alternatives and picture transcripts**

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case	Evidence	What it shows
Rain may start	forecast output	prediction
Dark clouds	observed condition	evidence
Rain gauge recorded water	later measurement	fact for evaluation
Animal label from model	unverified output	prediction, not fact

Repair two tempting claims

Repair:

Repair:

A closer explanation

MAIN LESSON · SECTION 8

Practise: include unassessed cases in the report

> A missing or conflicting outcome is not an established success or failure.

Count A through E with the stated assessed denominator

Consider only trials A through E. Three have established outcomes: A, B and C. Two of those three **prediction**s match and one does not. D and E are **unassessed**.

Report two matches out of three assessed trials, with two more trials **unassessed**. Do not report four successes by treating the unknowns as correct, or two out of five as if every trial had a known outcome.

> **Write a results record for A through E. Include each prediction, its observed category or unresolved evidence, its comparison status, and a summary that names both assessed and unassessed counts. Explain why D and E cannot be counted as correct**

Limit the result to the assessed group

The missing cases might be different from the checked cases. The result describes the assessed group; it does not settle performance on all five or on future trials.

Collect the missing **evidence** where possible, document what remains unknown, and keep the original model version visible. Week 23 will develop performance fractions more fully.

Use W2 to check the observation boundary

Use workbook W2 for the connected example "**Prediction**s and **observation**s". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Challenge “correct execution proves prediction correctness”

- Use the worked case for **Prediction**s and **observation**s. Proposed explanation: “Running the rule correctly proves the **prediction** correct”. Decide whether this explanation is supported.

  Point to the step or supplied **fact** that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Return to the two drying fields

Which field changes the **observation** label, and which field changes the model **prediction**?

> **My explanation using a pictured step and the given facts**

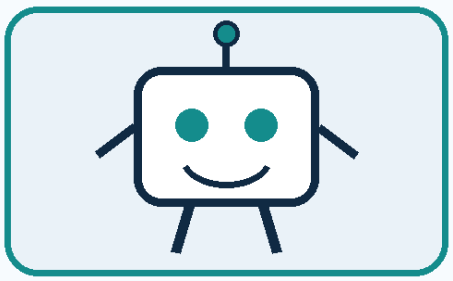
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: A **prediction** is a fallible **output** supported by clues.

Use a fresh example to explain **prediction**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: **Evidence** and **prediction** are two words for the same thing.

Invent one edge case and name the safe response.

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

Practise with less help: Check a boundary without changing the rule

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Investigate prediction and evidence records

> Use four cases with a match, mismatch, missing observation and conflicting observations.

The model, outcome policy and inputs

Use fictional CartC records. The model accepts whole push scores 1–5, predicts FAR at score 3 or more and NEAR otherwise. A later measured travel distance gives observed FAR at 5 floor units or more and NEAR otherwise. Trial inputs: A=2, B=3, C=4, D=5. Save every prediction with its trial ID before opening the observation cards below.

Save predictions, then compare the evidence

1. After saving predictions, reveal the observation cards. For each trial, distinguish its predicted category, supplied measurement evidence, observed category if established, and match/mismatch/unassessed status. Report matches out of assessed cases and name the unassessed count. Preserve conflicting reports and request the matching original trial record.

▼ Open the observation cards after predicting

Supplied observation cards for this paper exercise: A travelled 4 units; B travelled 4 units; C has no distance record; D has conflicting records of 4 and 6 units. These are fictional records, not measurements you have performed.

Keep comparison statuses and report limits

Keep a normal case and an edge case clearly separated.

Record what happened—even when it does not match your **prediction**.

Explain the result, one limitation, and the next useful test.

Case	Prediction	Observed result	What it means
Normal			
Edge			

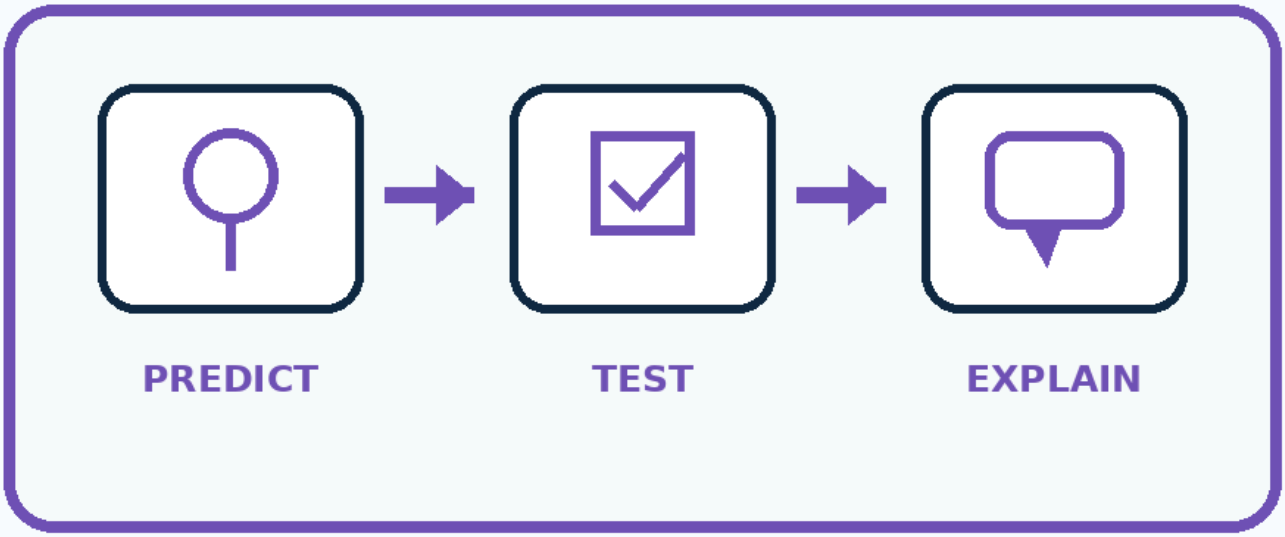
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

EXPLORER LAB

Prediction-Evidence-Fact Sort

Predict first, test honestly, and explain what the **evidence** means.



State your **prediction** before the result is known.

Complete the task: Separate **prediction**, **evidence**, and **fact** in four scenarios.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: compare the new Dry1 trials

> Apply the new score threshold and strict drying-time outcome rule, retaining missing evidence.

The new rules and records

In a fictional paper test, Dry1 predicts slow for a moisture score of at least 7, quick otherwise. Allowed scores are whole numbers 1 through 10.

The observed outcome is slow if drying time is greater than 10 minutes, quick otherwise. Records are P: score 6, time 12; Q: score 7, time 10; R: score 9, time 14; S: score 5, time missing.

Record all **prediction**s, observed categories and comparison statuses. Summarise matches out of assessed cases, name the **unassessed** count, and limit the conclusion.

> **My independent explanation, trace and limits**

Keep one first attempt

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Transfer the check to drying predictions

Week 17 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A model predicts rain tomorrow. Tomorrow is dry. Was the **prediction** an **observation** of tomorrow? Explain what new **evidence** settles the result.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Label the sorter's suggested bin as a **prediction** until a human confirms the final placement.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

On Monday a team predicts 'rain on Tuesday'. On Tuesday they count a damp playground as success, although sprinklers ran. Write an **observation** rule that could have been agreed before Tuesday.

Explain the mechanism, evidence and one limit.

Use feedback to improve

MAIN LESSON · SECTION 11

Keep sorter predictions before checking placement

> Retain disagreements and the evidence used to investigate them.

This week’s project decision

THIS WEEK'S CHECKPOINT Label the sorter's suggested bin as a **prediction** until a human confirms the final placement.

In sorter trials, record predicted categories before collecting reference **evidence**, then retain disagreements for investigation.

Plan IDs, predictions, checks and review

- What real classroom need does this serve?
- What information is necessary—and what should stay private?
- What happens inside the **process**?
- What normal and edge cases will you test?

- When should it say not sure or ask a person?

Keep the comparison record independent of the prediction

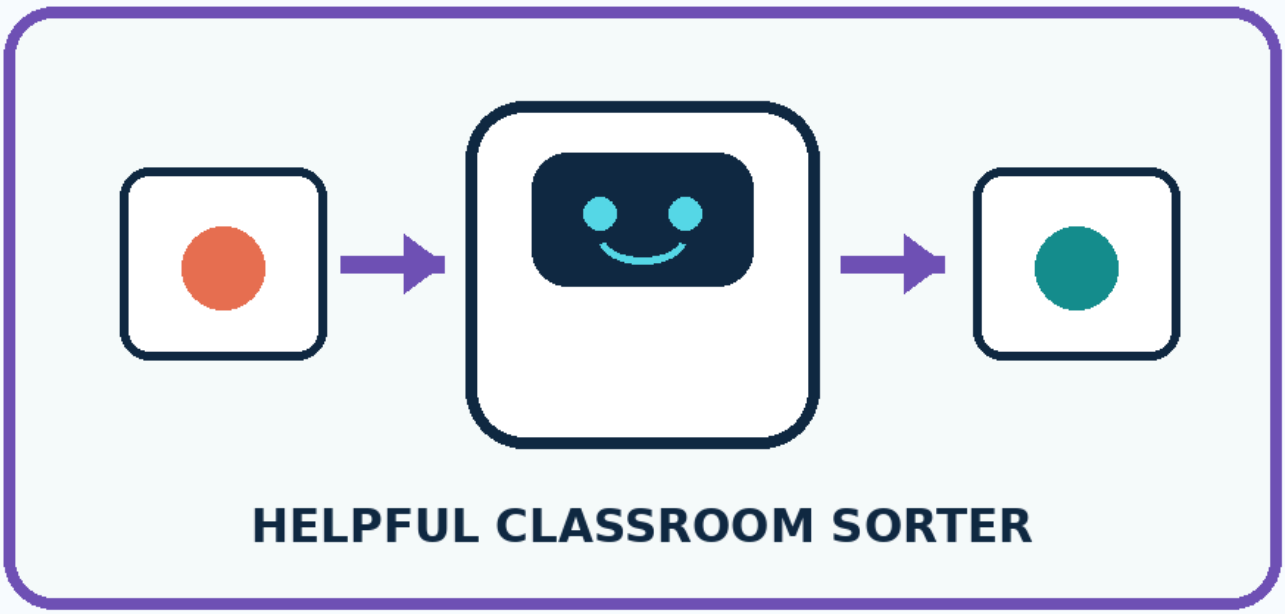
RESPONSIBLE DESIGN Do not copy a model **prediction** into the answer column and later call it **ground truth**.

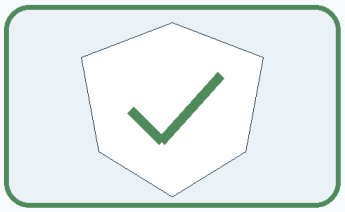
▼ **Earlier wording, alternatives and picture transcripts**

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.





My project decision:

Evidence for your project

MAIN LESSON · SECTION 16

Finish: report the assessed and unresolved evidence

> Close your notes and explain the model output, outcome rule and limits of the comparison.

Your exit explanation

- Close your notes. Explain this goal in your own words: Separate a recorded **prediction** from the **observation** used to check it

My deciding step, evidence and one remaining question

Choose support and revisit later

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	Meaning	My example
prediction	estimate about an unknown outcome	
evidence	information that supports or challenges a claim	
fact	claim established by direct or reliable checking	
ground truth	reference outcome used to evaluate a prediction	

Terms for the connected explanation

Word	Meaning
Observation	A recorded measurement or event used as evidence
Outcome rule	The agreed method for turning observed evidence into the category being checked
Mismatch	A prediction that differs from an established comparison outcome

Word	Meaning
Unassessed	A case whose prediction has not yet been judged against adequate outcome evidence

Repair a claim using a word

- CLAIM TO REPAIR A confident **prediction** is a **fact**.
- Use one vocabulary word and one piece of **evidence** to repair it:

▼ Earlier wording, alternatives and picture transcripts

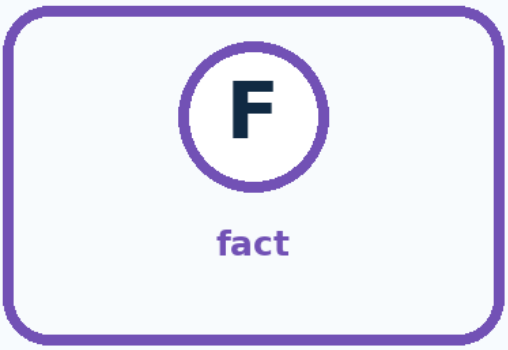
VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.









Misconception detective

WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

▼ Choose a stronger thinking challenge

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

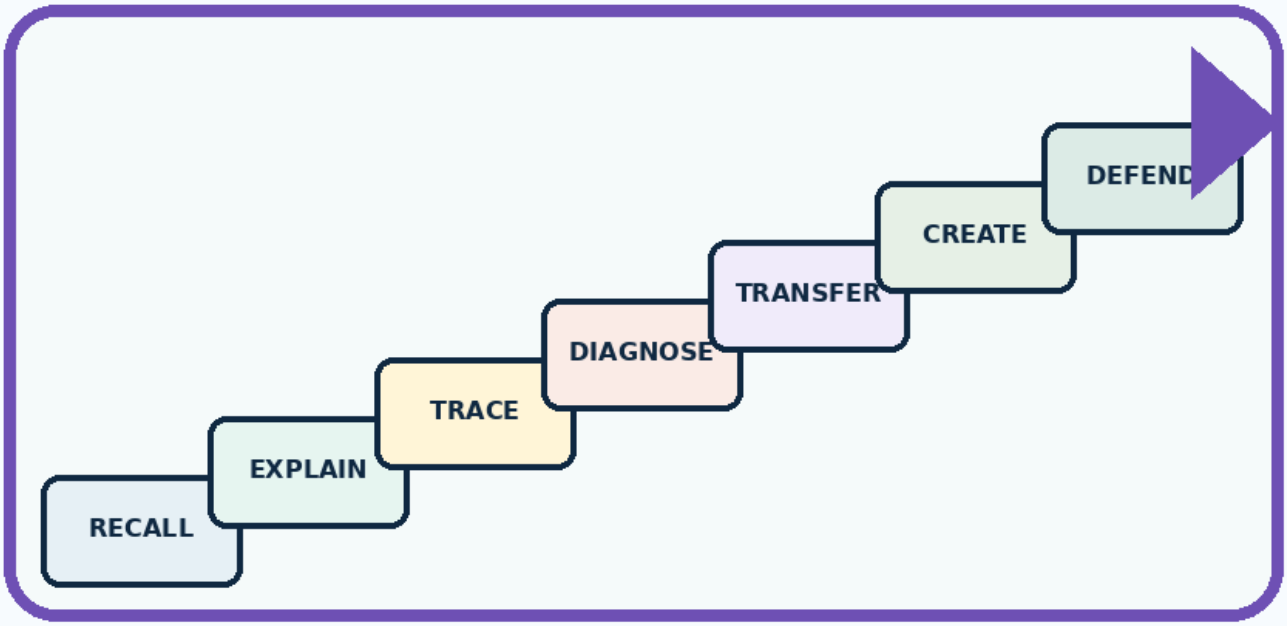
- RECALL Define **prediction**.
- EXPLAIN A **prediction** is a fallible **output** supported by clues.
- TRACE Show the **input**, **process**, and **output**.
- DIAGNOSE Repair: A confident **prediction** is a **fact**.
- TRANSFER Apply the idea to a new home or classroom case.
- CREATE Design one revealing test.
- DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

▼ Earlier wording, alternatives and picture transcripts

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

▼ **Reference: check your explanation**

REFERENCE · SECTION 12

Reference: check your explanation

> Use after practice. Transfer the check to drying predictions is the selected independent case.

Check the evidence and its limit

- ☐ Name or draw the fresh case.
- ☐ Trace **input**, **process**, and **output**.

- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Record support

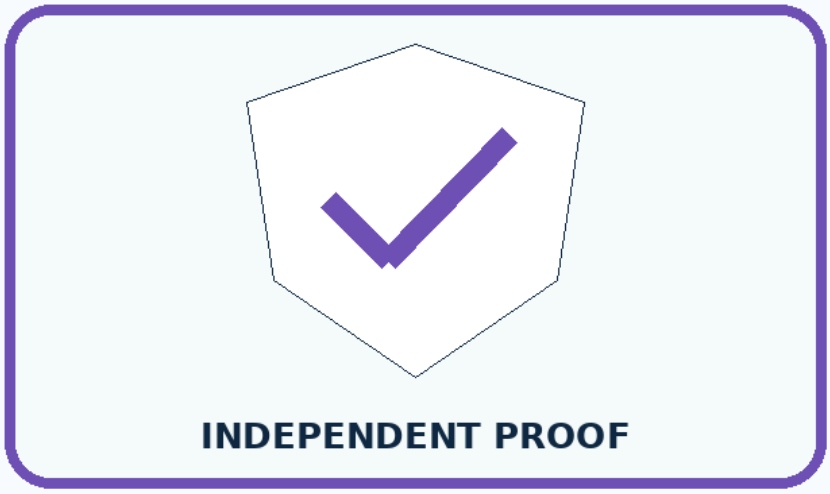
My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

▼ Earlier wording, alternatives and picture transcripts

MASTERY

Checklist for explaining this week's idea

Use after practice. Transfer the check to drying predictions is the selected independent case.



EXPLANATION CHECK Keep **prediction**s and later observations separate, apply the same-case outcome rule, preserve unresolved **evidence** and state the assessed denominator.

Independent **evidence**:

SELF-CHECK Mark model or rule **output**s as **prediction**s, observable clues as **evidence**, and independently checked outcomes as **fact**s.

A **prediction** becomes evaluated against a **fact**; it does not become a **fact** merely because time passes.

NEXT BRIDGE Carry today's **evidence** habit into: Confidence Is Not Correctness.

▼ **Extra practice: A prediction becomes checkable evidence**

REFERENCE · SECTION 13

Extra practice: A prediction becomes checkable evidence

> Try the question before opening its worked explanation.

Read the supplied case and question

A weather helper predicts tomorrow's rain.

Is this a **fact** about tomorrow? How can it be checked?

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

It is a **prediction**. Record it now and compare with a predefined rain **observation** tomorrow.
Useful **prediction**s are not guarantees.

Evidence or stage	Value or result
Today	Record tomorrow's rain prediction
Tomorrow	Observe using an agreed rain rule
Compare	Prediction versus observation

▼ Earlier wording, alternatives and picture transcripts

Week 17 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

A prediction becomes checkable evidence

A weather helper predicts tomorrow's rain.

Is this a **fact** about tomorrow? How can it be checked?

It is a **prediction**. Record it now and compare with a predefined rain **observation** tomorrow.
Useful **prediction**s are not guarantees.

▼ Reference: original lesson checklist

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Separate a recorded prediction from the observation used to check it	Explain the deciding step, show a trace or test, and state one limit.
Compare matching cases using a stated outcome rule and preserve unresolved evidence	Explain the deciding step, show a trace or test, and state one limit.
Report checked results without counting missing observations as successes or failures	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: A fictional drying trial uses a fixed rule and later measures what actually happened.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional drying trial uses a fixed rule and later measures what actually happened.
- Which labelled step matches this detail? **Prediction** quick disagrees with **observation** slow.
- 1. Predict
- 2. Compare **evidence**
- 3. Observe later
- 4. **Prediction input**
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ Check the first explanation

After your attempt, compare with the supported explanation: Compare **evidence Prediction** quick disagrees with **observation** slow. The moisture score generates a **prediction**; it is not the observed drying time.

The later measurement supplies separate **evidence** under the stated **observation** policy. Save both values and their units. Altering the recorded time to match D would destroy the evaluation rather than repair the model.

Changed-case question

- Changed-case practice: Keep score 4 but measure exactly 8 minutes. Which conclusion is supported by the supplied **fact**s?
- 1. The observed label becomes quick and agrees with the **prediction**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those **fact**s.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ Check the changed case

After your attempt, compare with the supported explanation: The observed label becomes quick and agrees with the **prediction**. Above 8 is strict: equality is quick under this policy.

Repair and save your work

Review Boundaries and 'Not Sure' (ai-1/16). Goal: Identify a specific source of uncertainty and explain how it affects a **prediction**. Short practice: Which original block needs a better width before accepting or rejecting it?

What should the reviewer do?. Return to this week and explain what you changed.

Review Build a Human Classifier (ai-1/14). Goal: Construct a classifier with a stated **input** format, selected rule and recorded **output**.

Short practice: Trace 0 cm and explain why clarification is different from predicting short.. Return to this week and explain what you changed.

Review Repeating Efficiently (ai-1/11). Goal: Trace a loop by testing its condition before every repetition. Short practice: Trace a fresh start of 9. How many additions occur, and must the report equal 10?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

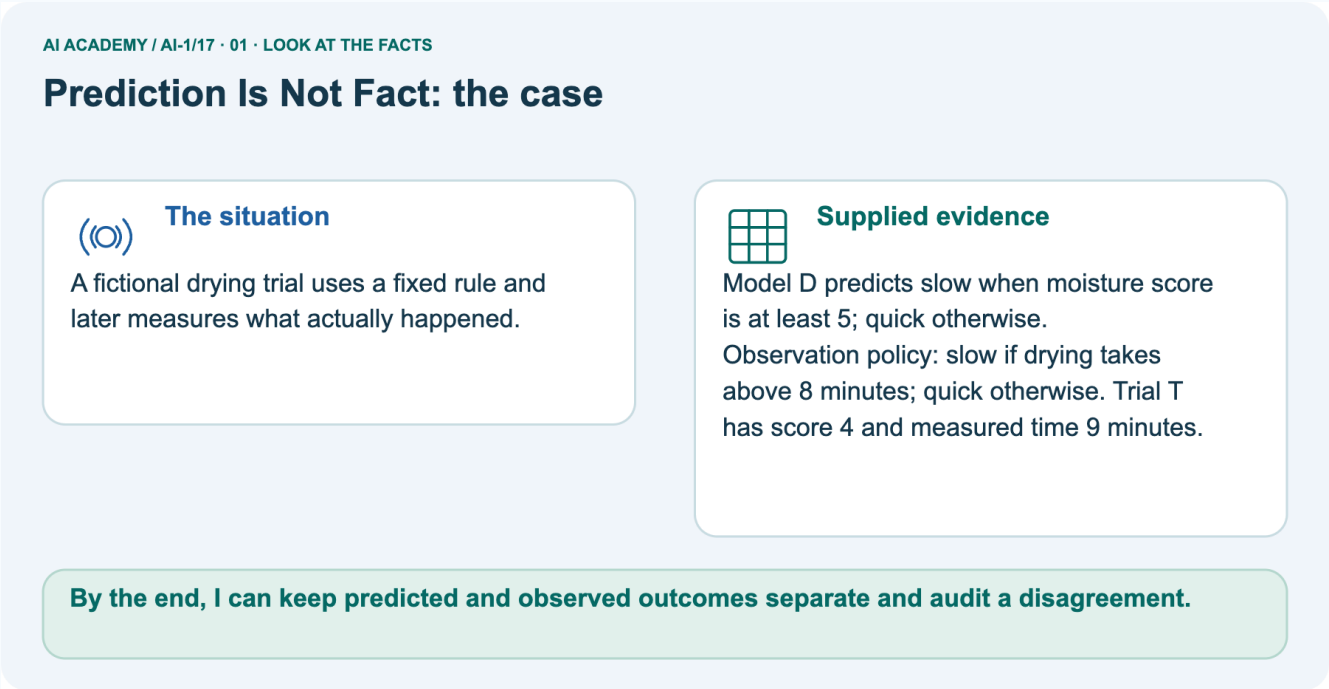
Keep your completed work

▼ Reference: the six-picture journey

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Prediction Is Not Fact: the case



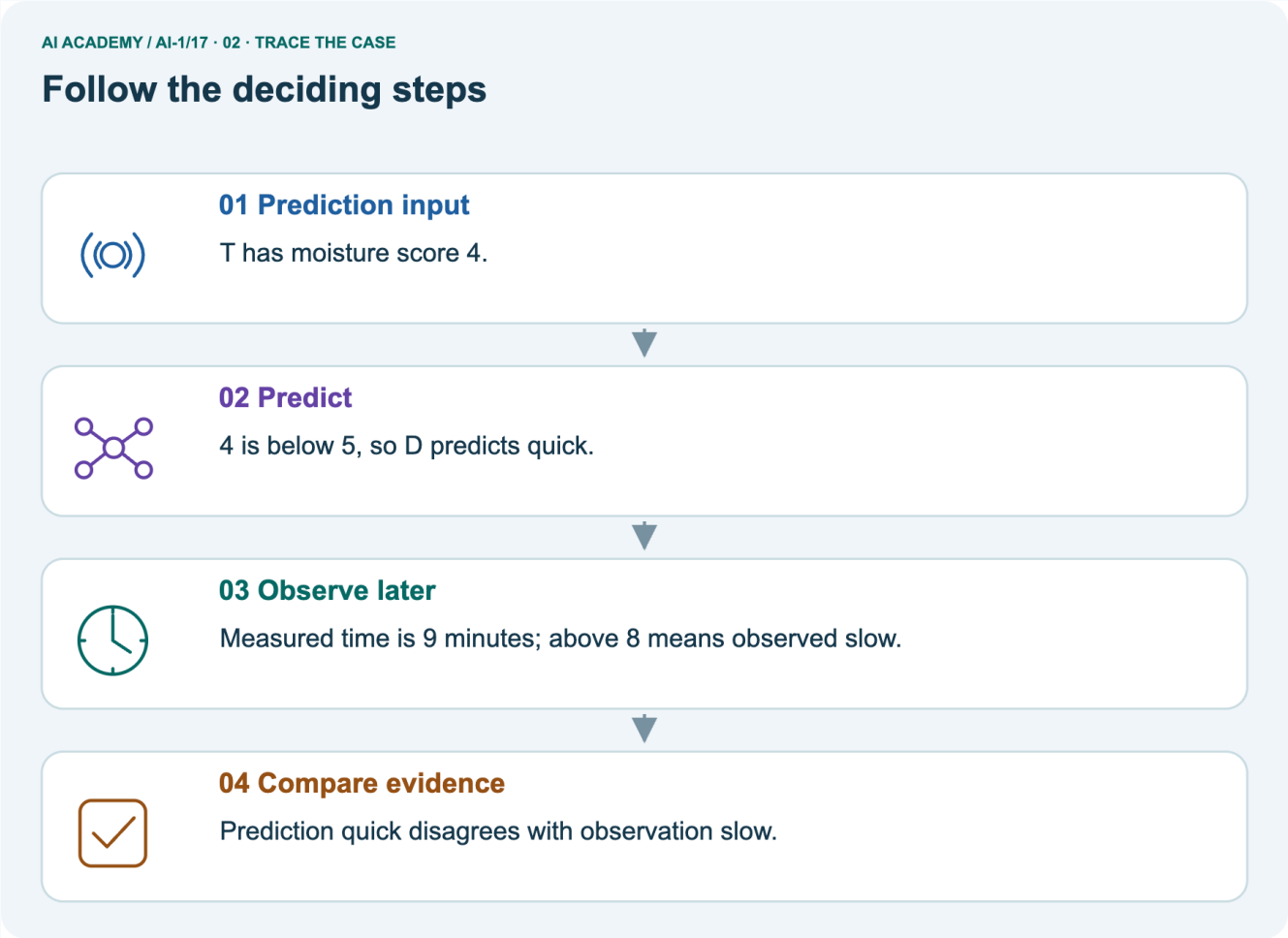
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional drying trial uses a fixed rule and later measures what actually happened. Supplied **fact**s: Model D predicts slow when moisture score is at least 5; quick otherwise.

Observation policy: slow if drying takes above 8 minutes; quick otherwise. Trial T has score 4 and measured time 9 minutes.

Point to the given **fact** that matters most. Explain why it matters.

02. Follow the deciding steps

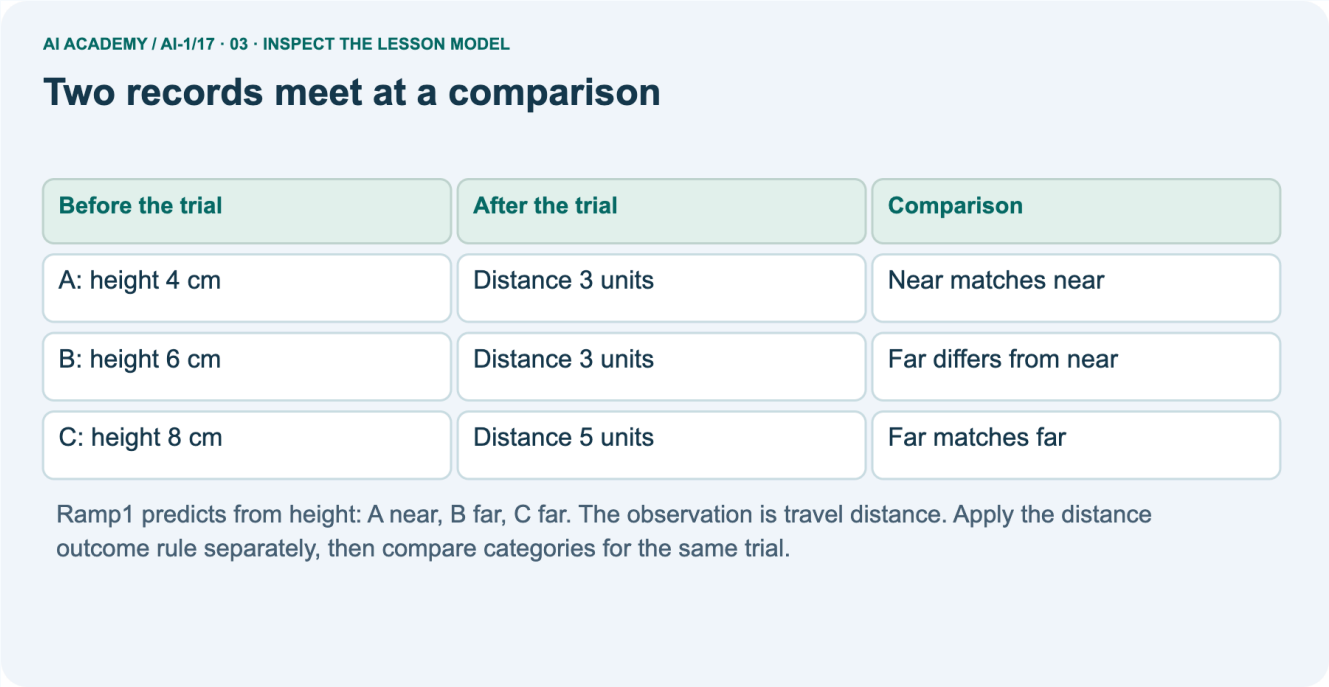


Follow the labelled picture with the matching explanation. Open the vector diagram

Prediction input: T has moisture score 4. → Predict: 4 is below 5, so D predicts quick. → Observe later: Measured time is 9 minutes; above 8 means observed slow. → Compare **evidence**: **Prediction** quick disagrees with **observation** slow.

Point to each numbered stage and explain which recorded **input** it uses.

03. Two records meet at a comparison

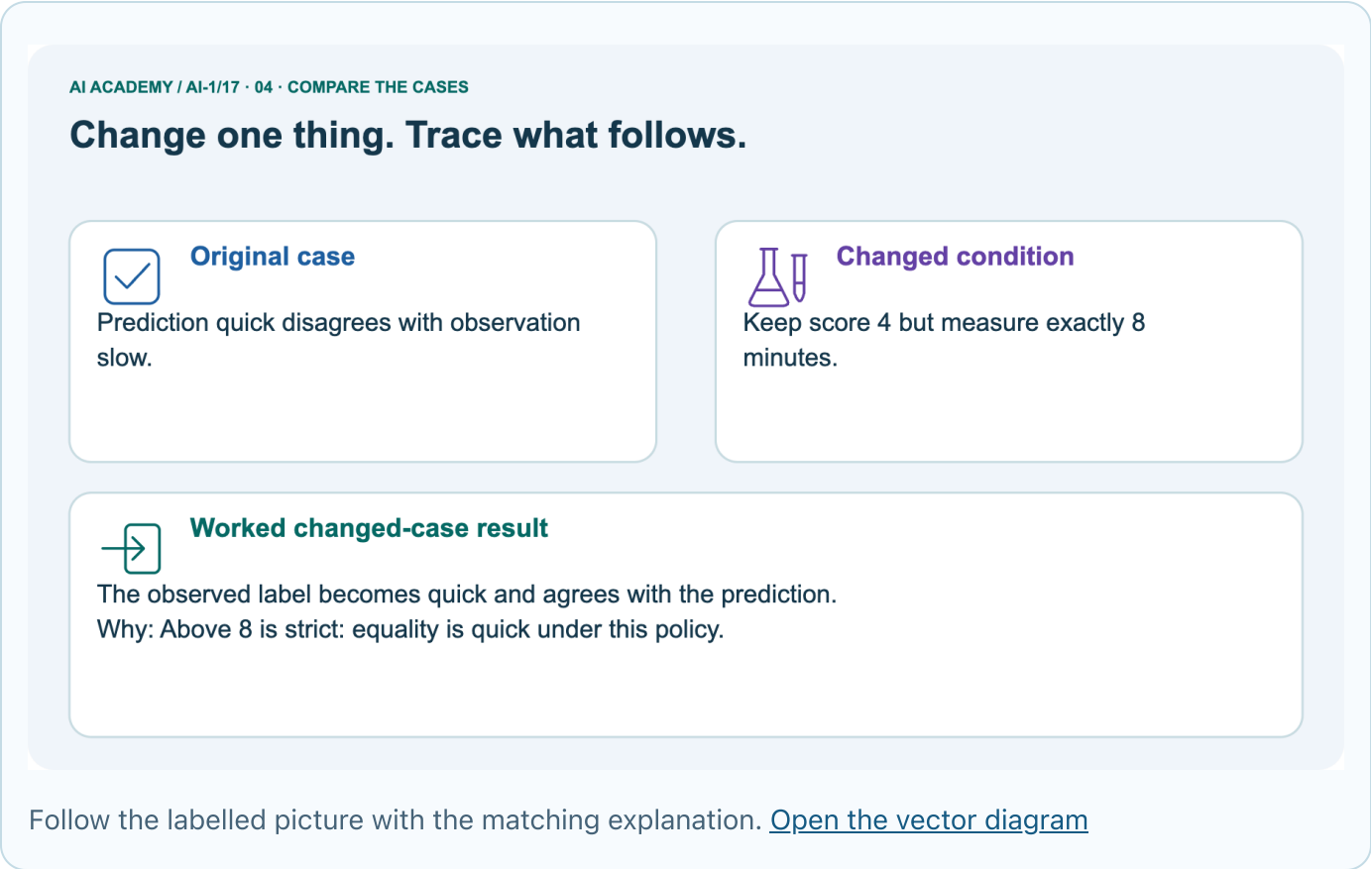


Follow the labelled picture with the matching explanation. Open the vector diagram

Two records meet at a comparison. A: height 4 cm | Distance 3 units | Near matches near. B: height 6 cm | Distance 3 units | Far differs from near. C: height 8 cm | Distance 5 units | Far matches far

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



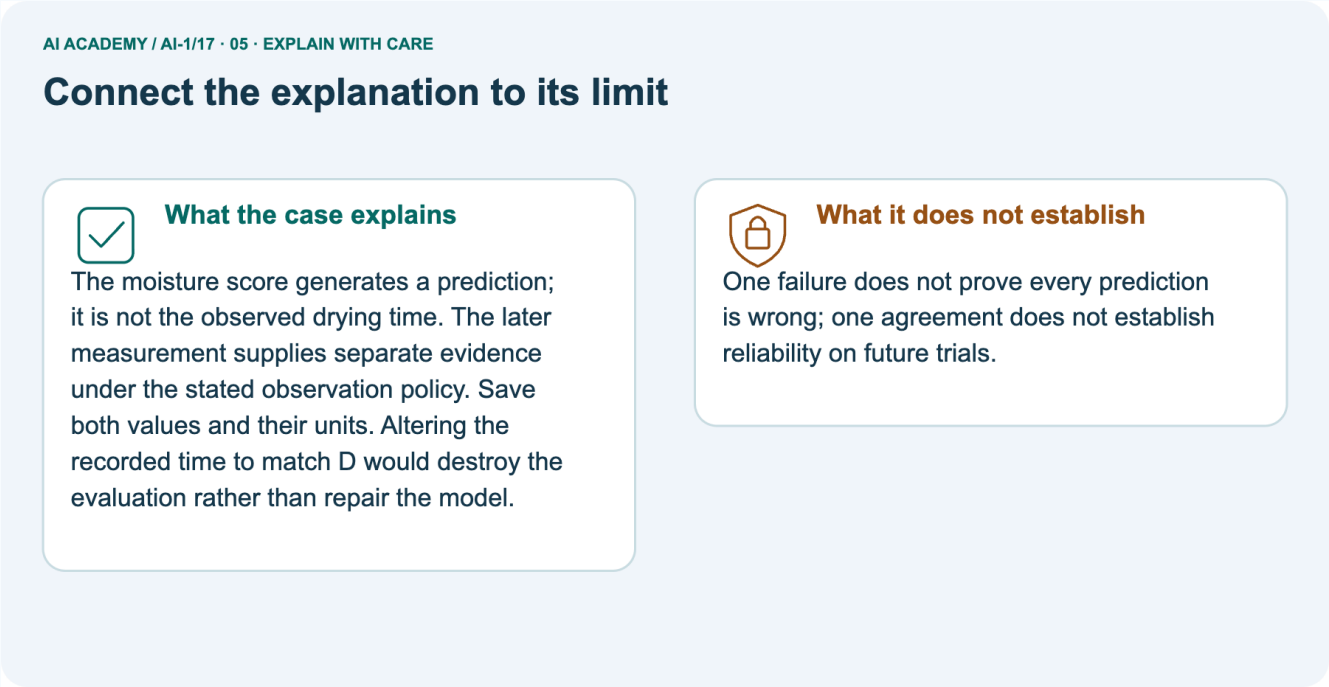
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: **Prediction** quick disagrees with **observation** slow. Changed condition: Keep score 4 but measure exactly 8 minutes. Expected result: The observed label becomes quick and agrees with the **prediction**.

Reason: Above 8 is strict: equality is quick under this policy.

Predict before checking the worked changed-case result. Name what changed and what the supplied **fact**s allow you to conclude.

05. Connect the explanation to its limit



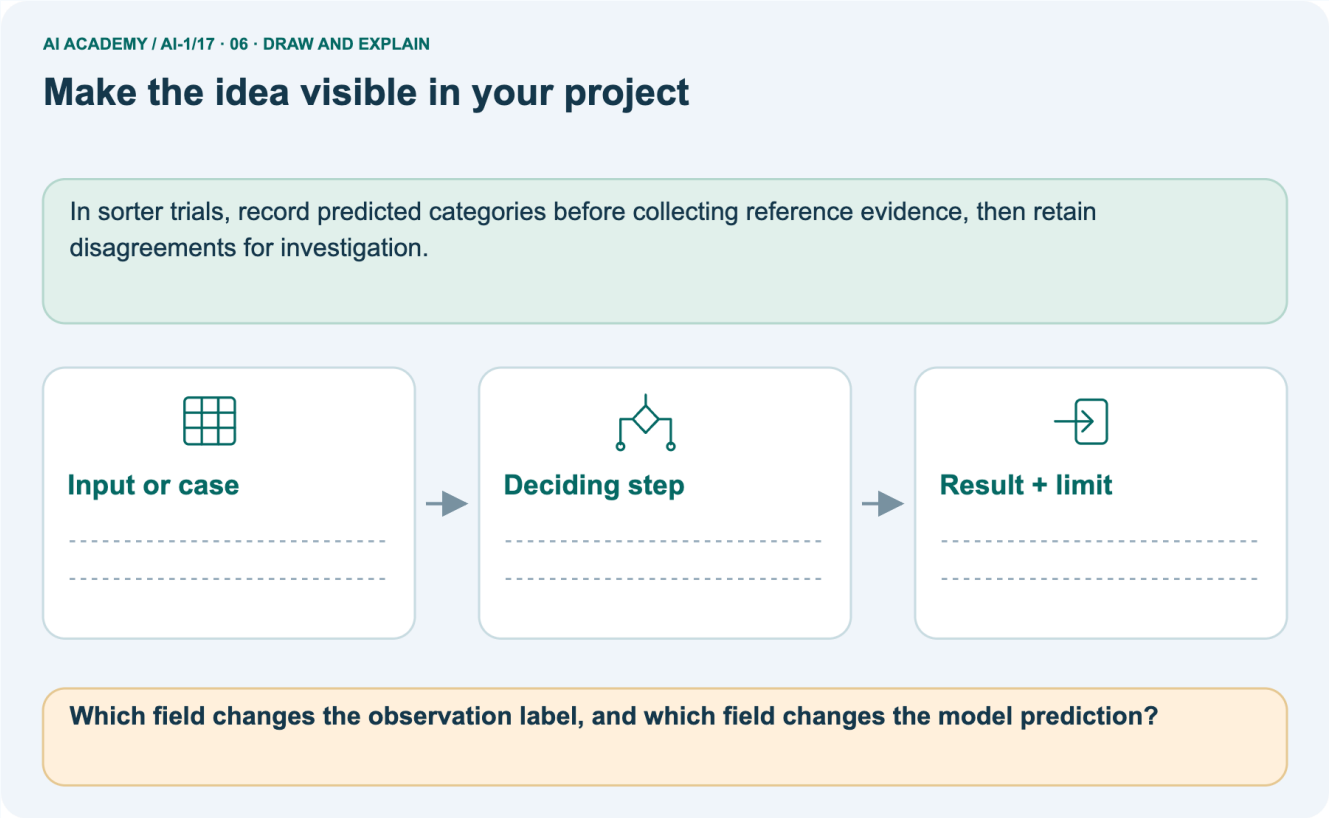
Follow the labelled picture with the matching explanation. Open the vector diagram

The moisture score generates a **prediction**; it is not the observed drying time. The later measurement supplies separate **evidence** under the stated **observation** policy. Save both values and their units.

Altering the recorded time to match D would destroy the evaluation rather than repair the model. Limit: One failure does not prove every **prediction** is wrong; one agreement does not establish reliability on future trials.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

In sorter trials, record predicted categories before collecting reference **evidence**, then retain disagreements for investigation. Draw your **input**, deciding step, result and limit.

Supported question: Which field changes the **observation** label, and which field changes the model **prediction**?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Prediction Is Not Fact: the case

02. Follow the deciding steps

03. Two records meet at a comparison

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.